

B. Tech. Examination, 2024-25

(Odd Semester)

DIGITAL LOGIC DESIGN

Time : Three Hours/

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION - A

1. Attempt all parts of the following : $8 \times 1 = 8$
- (a) What is gray code? Give the advantage of gray code over binary code.
- (b) Find the standard SoP form of the following function :
- $$f(A, B, C, D) = \bar{A} + BC\bar{D} + A\bar{C}$$
- (c) What do you mean by sign-magnitude representation?

[P. T. O.]

- (d) What are the different hazards in combinational circuits?
- (e) Implement the following function using a multiplexer :

$$Y(A, B, C) = \sum m(1, 2, 5, 6)$$

- (f) Explain the difference between asynchronous and synchronous sequential circuits.
- (g) What are programmable logic devices?
- (h) What is the significance of state assignment?

SECTION - B

2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) Minimize the following Boolean function using K-map :

$$(i) \quad Y(A, B, C, D) = \sum m(1, 3, 5, 8, 9, 11, 15) + d(2, 13)$$

$$(i) \quad Y(A, B, C, D) = \pi_M(1, 2, 3, 8, 9, 10, 11, 14) \cdot d(7, 15)$$

- (b) Implement the following combinational logic circuits using multiplexer :
- (i) Half adder
 - (ii) Half subtractor
- (c) Draw the logic circuit of J-K flip-flop using D-flip flop.
- (d) Design a Mod-5 synchronous counter using T-flip flop.

SECTION - C

Note :- Attempt all questions. Attempt any two parts from each questions. 8×5=40

3. (a) Explain with example, how Hamming code is useful for detecting and correcting error in digital data transmission?
- (b) Realize the following expression using EX-NOR gates :

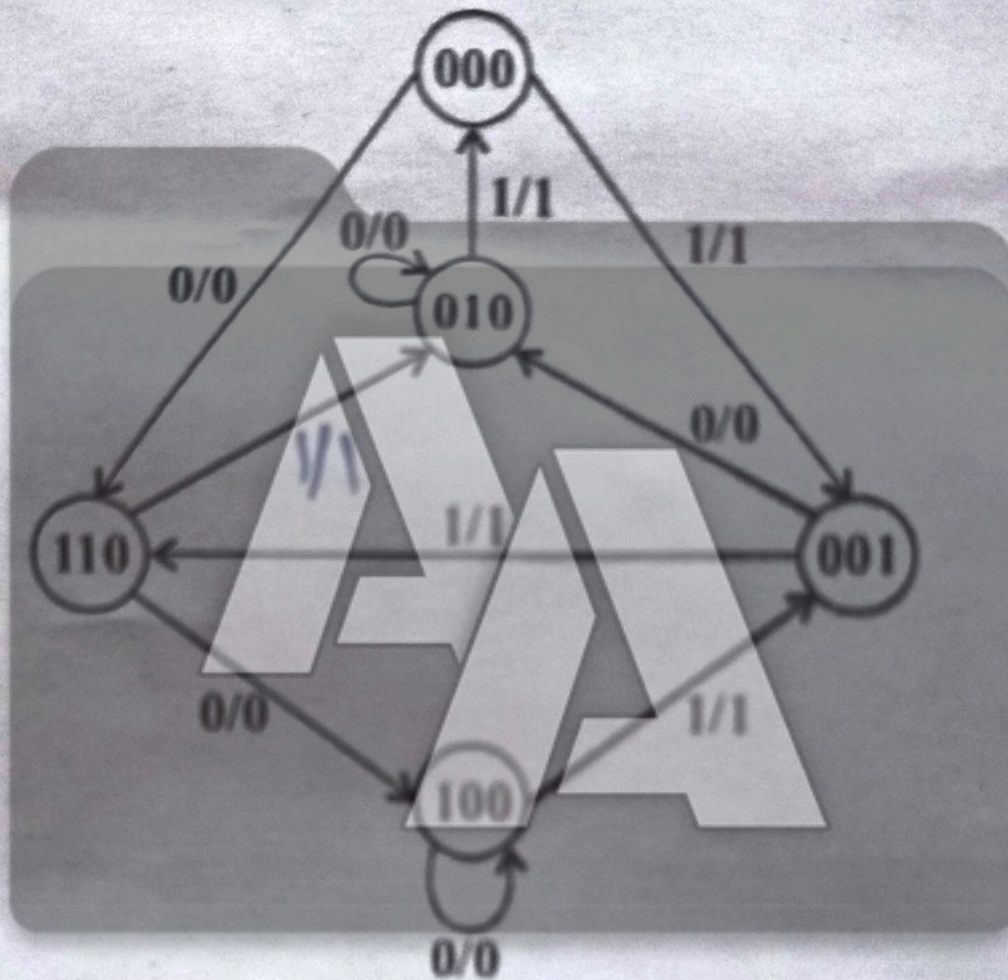
$$f = \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + AB\bar{C}D$$

- (c) Simplify the following four-variable Boolean function using Quine-McCluskey method :

$$Y(A, B, C, D) = \sum m(2, 4, 5, 9, 12, 13)$$

4. (a) Design a 4-bit binary-to-gray code converter using truth table, K-map and logic circuits.
- (b) Describe the difference between PLA and PAL.
- (c) Design a 4-bit odd parity generator using suitable gates.
5. (a) What is race-around condition? How does it get eliminated in a master-slave JK-flip flop? Explain.
- (b) Design a Mod-9 ripple counter using T-flip-flop.
- (c) Design a sequential generator using JK-flip-flop to generate the sequence :
- $$0 \rightarrow 2 \rightarrow 4 \rightarrow 5 \rightarrow 1 \rightarrow 7 \rightarrow 6$$
6. (a) Describe the design procedure for asynchronous sequential circuits.
- (b) Explain the methods to eliminate static hazards in an asynchronous circuit. Also define races in asynchronous sequential circuits.

- (c) Implement the following state diagram using D-flip flop :



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